

LOGISTICS PERFORMANCE ANALYSIS PROJECT

Power Pivot for Beginners

BUSINESS SCENARIO

You are a **Data Analyst at FastShip Logistics**, a mid-sized shipping company that partners with multiple carriers (UPS, FedEx, DHL, USPS, Amazon Logistics, OnTrac, LaserShip) to deliver packages across the United States.

Your company operates **10 major warehouse locations**:

- **East Coast:** Miami (MIA), Boston (BOS), New York (NYC), Atlanta (ATL)
- **Central:** Chicago (CHI), Houston (HOU), Denver (DEN)
- **West Coast:** Los Angeles (LA), San Francisco (SF), Seattle (SEA)

THE SITUATION

Your Operations Manager, **Sarah Chen**, has noticed inconsistent delivery performance and rising costs over the past year (2023). The executive team is concerned about:

1. **Rising shipping costs** - Are we spending too much on certain routes or carriers?
2. **Delivery delays** - Which carriers or warehouses are causing delays?
3. **Lost packages** - Why are some shipments getting lost or returned?
4. **Operational efficiency** - Can we optimize our carrier partnerships?

Sarah has given you **2,000 shipment records** from 2023 and asked you to create an **interactive dashboard** that the operations team can use to monitor performance and make data-driven decisions.

YOUR MISSION

Create a Power Pivot analysis and dashboard that answers these critical business questions:

COST ANALYSIS

1. What is our **total shipping cost** for 2023?
2. What is the **average cost per shipment**?

3. Which **carrier** is most expensive on average?
4. Which **warehouse** has the highest total shipping costs?
5. What is the **cost per mile** for each carrier? (to measure efficiency)

DELIVERY PERFORMANCE

6. What percentage of shipments were **delivered on time**?
7. Which carrier has the **best delivery success rate**?
8. Which carrier has the **worst performance** (most delays/lost packages)?
9. What is the **average transit time** across all shipments?
10. Which **warehouse** has the longest average transit times?

OPERATIONAL INSIGHTS

11. What is our **total shipped weight** for the year?
12. What is the **average package weight**?
13. Which **destination city** receives the most shipments?
14. Which **carrier** handles the most volume (number of shipments)?
15. Is there a relationship between **distance** and **transit time**?

PROBLEM AREAS

16. How many shipments were **Lost** or **Returned**?
17. What percentage of our shipments have **issues** (Delayed, Lost, Returned)?
18. Which carrier has the **most lost packages**?
19. Which warehouse has the **highest problem rate**?



DELIVERABLES

You need to create:

1. **Power Pivot Data Model**
 - Import the shipment data
 - Create a Calendar table for time-based analysis

- Build necessary relationships
 - 2. **DAX Measures** (10-15 key metrics)
 - Total costs, counts, averages
 - Performance percentages
 - Calculated rates and ratios
 - 3. **Interactive Dashboard** (1-2 pages)
 - **Executive Summary:** High-level KPIs and trends
 - **Carrier Performance:** Compare carriers on cost and delivery
 - **Warehouse Analysis:** Identify problem locations
 - **Problem Tracking:** Monitor delays, losses, and returns
 - 4. **Insights Report**
 - Answer all 19 business questions
 - Provide 3-5 actionable recommendations
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LEARNING OBJECTIVES

By completing this project, you will learn to:

- ✓ Import data into Power Pivot
 - ✓ Create and manage a Data Model
 - ✓ Build a Calendar dimension table
 - ✓ Write basic DAX measures (SUM, AVERAGE, COUNT)
 - ✓ Create calculated measures (percentages, ratios)
 - ✓ Use DAX functions (CALCULATE, FILTER, RELATED)
 - ✓ Build pivot tables from the Data Model
 - ✓ Create interactive dashboards with slicers
 - ✓ Analyze data to answer business questions
 - ✓ Present insights professionally
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YOUR DATA

You have been provided with **Shipment_Data.xlsx** containing 2,000 shipment records with the following fields:

Field	Description
Shipment_ID	Unique shipment identifier (SH10000 - SH12000)
Origin_Warehouse	Warehouse where shipment originated
Destination	Destination city
Carrier	Shipping carrier used
Shipment_Date	Date package was shipped
Delivery_Date	Date package was delivered (blank if not delivered)
Weight_kg	Package weight in kilograms
Cost	Shipping cost in USD
Status	Delivery status (Delivered, Delayed, In Transit, Lost, Returned)
Distance_miles	Distance traveled in miles
Transit_Days	Number of days in transit

PROJECT TIMELINE

Session 1 (2-3 hours): Data Model Setup & Basic Measures

- Import data and explore structure
- Create Calendar table
- Build 8-10 core DAX measures
- Create basic pivot tables

Session 2 (2-3 hours): Advanced Analysis & Dashboard

- Create advanced measures (percentages, ratios)
- Build interactive dashboard
- Answer business questions

- Write insights report

Total Time: 4-6 hours

SUCCESS CRITERIA

Your analysis will be considered successful if you:

- ✓ Create a working Power Pivot data model
 - ✓ Build all required DAX measures accurately
 - ✓ Answer all 19 business questions with data
 - ✓ Create a professional, interactive dashboard
 - ✓ Provide at least 3 actionable recommendations
 - ✓ Present insights clearly to non-technical stakeholders
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READY TO BEGIN?

Sarah Chen is counting on you to deliver insights that will help FastShip Logistics optimize their operations and reduce costs.

Let's dive into Power Pivot and turn this data into decisions!

Note: This is a beginner-level project designed to teach Power Pivot fundamentals through a realistic business scenario. All data is simulated for educational purposes.